

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK SECURITY (CYS360)
(6TH SEMESTER, 3RD Year) LAB EXPERIMENT # 15

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Score:	<u> </u>	Date:	<u> </u>	Instructor's Signature:	<u> </u>

Lab # 15: To configure Intrusion Prevention System on Firewall

Objective: The objective of this lab is to configure Intrusion Prevention System on Firewall

Duration: 3 Hours

Expected Outcomes: By the end of this lab session, students will be able to:

1. Ensure an intranet user can access the intranet FTP server and the Internet web server (verify file transfer and HTTP fetch).
2. Configure and validate access controls so the FTP server serves both intranet and Internet users while protecting intranet-only resources.

Rubrics

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT
SCORE			

Note: Student performance will be assessed based on this rubric during lab sessions.

Networking Requirements

As shown in [Figure 1](#), an enterprise deploys a FW as the security gateway at the network border. In the networking:

- An intranet user can access the intranet FTP server and Internet web server.
- The intranet FTP server provides services for both intranet and Internet users.

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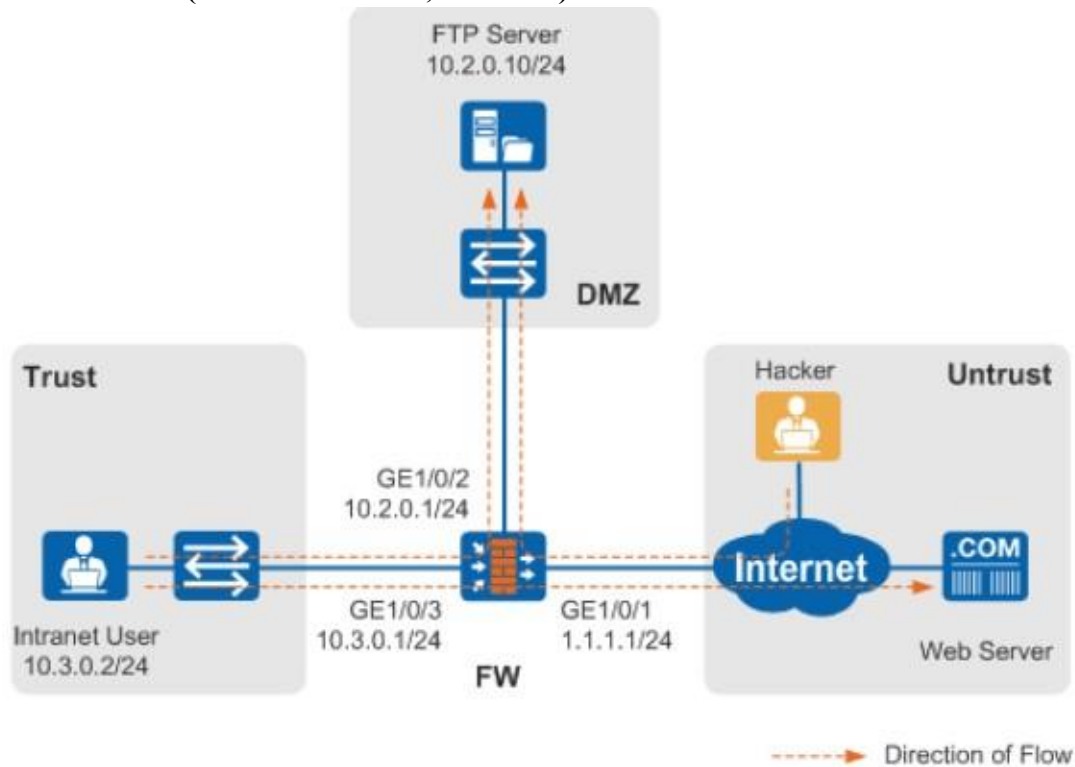


Figure 1 Intrusion prevention networking diagram

Note: You need to make the topology on eNSP by yourself. See Figure # 1 to make the topology on eNSP.

The enterprise wants to enable intrusion prevention on the FW to meet the following requirements:

- Defend against worms, Trojan horses, and botnet attacks.
- Protect intranet users.

Protect intranet users from attacks, such as an attacker launched from a website with malicious code, when the users access the Internet web server.

- Protect the intranet FTP server.

Prevent Internet and intranet users from launching attacks on the intranet FTP server.

An attack matching the signature with ID 74320 occurs frequently in logs and must be blocked.

Data Planning

Based on the previous enterprise requirements, the intrusion prevention information to be configured is as follows:

- Attacks on the enterprise include common worms, Trojan horses, and botnets, and the severity of these attacks in signatures is **High**.

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- Protect intranet users.
- Configure security policies for the direction from the Trust zone to the Untrust zone.
- Attacks are caused by intranet users' access to the Internet web server and the target is the intranet users acting as a client. Therefore, set the protocol to **HTTP**, object to **Client**, and severity to **High** for the signature filter.

The data planning for protecting intranet users is shown in [Figure 2](#).

Figure 2 Data planning for protecting intranet users

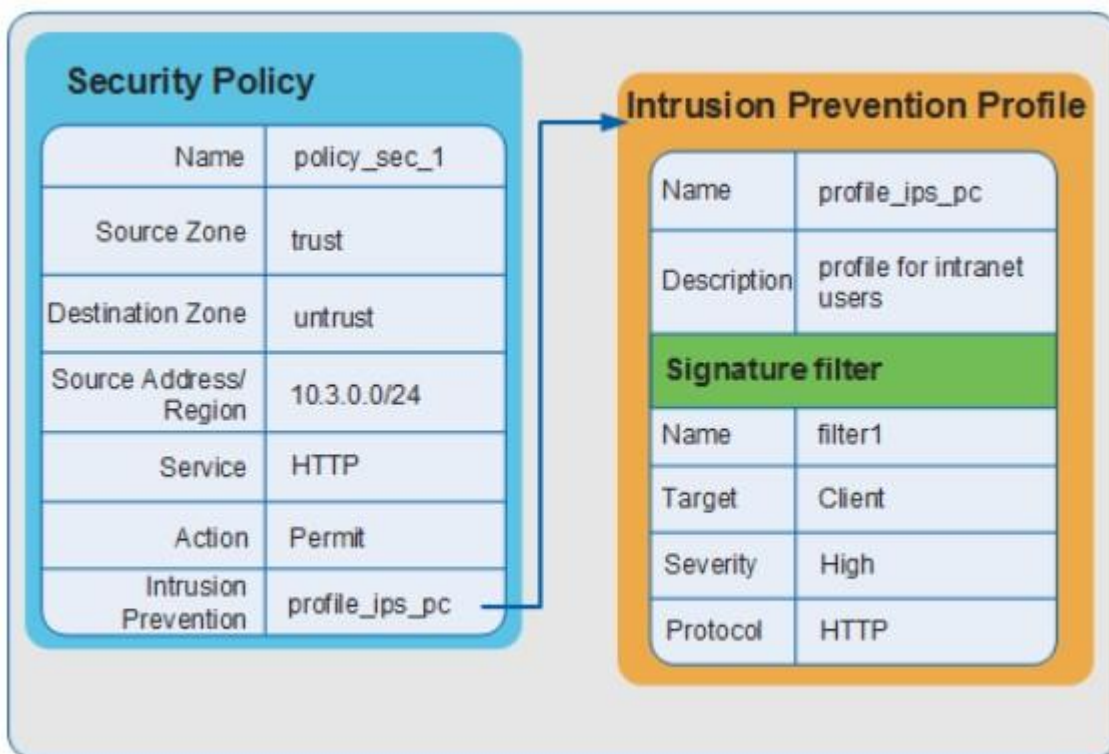


Figure 2 Data planning for protecting intranet users

Protect the intranet FTP server.

- Configure security policies for the directions from the Untrust zone to the DMZ zone and from the Trust zone to the DMZ zone.
- Attacks on the FTP server. Therefore, set the protocol to **FTP**, object to **Server**, and severity to **High** for the signature filter.
- Add the signature with ID 74320 to the exception signatures and set the action to **Block**.

The data planning for protecting intranet FTP server is shown in [Figure 3](#).

Figure 3 Data planning for protecting intranet FTP server

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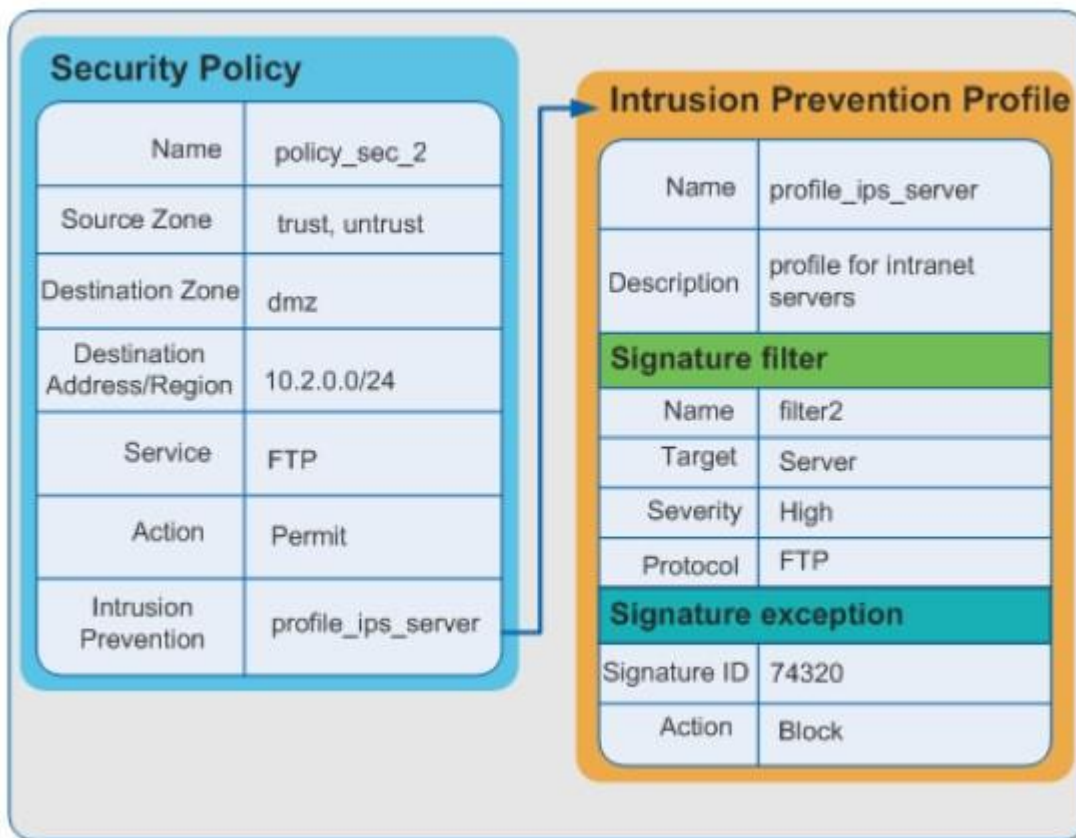


Figure 3 Data planning for protecting intranet FTP server

Configuration Roadmap

The configuration roadmap is as follows:

1. Set the interface IP addresses and add the interfaces to corresponding security zones as required.
2. Configure intrusion prevention profile **profile_ips_pc** to protect intranet users. Then configure a signature filter to meet the requirement.
3. Configure intrusion prevention profile **profile_ips_server** to protect intranet servers. Then configure a signature filter and add signatures as exceptions.
4. Create security policy **policy_sec_1** and reference profile **profile_ips_pc** to protect intranet users from Internet attacks.
5. Create security policy **policy_sec_2** and reference profile **profile_ips_server** to protect intranet servers from intranet and Internet attacks.

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Procedure

1. Set an IP address for each interface, assign interfaces to security zones, and complete basic parameter settings.

```
[FW] interface GigabitEthernet 1/0/1
[FW-GigabitEthernet1/0/1] ip address 1.1.1.1 255.255.255.0
[FW-GigabitEthernet1/0/1] quit
[FW] interface GigabitEthernet 1/0/2
[FW-GigabitEthernet1/0/2] ip address 10.2.0.1 255.255.255.0
[FW-GigabitEthernet1/0/2] quit
[FW] interface GigabitEthernet 1/0/3
[FW-GigabitEthernet1/0/3] ip address 10.3.0.1 255.255.255.0
[FW-GigabitEthernet1/0/3] quit
[FW] firewall zone trust
[FW-zone-trust] add interface GigabitEthernet 1/0/3
[FW-zone-trust] quit
[FW] firewall zone dmz
[FW-zone-dmz] add interface GigabitEthernet 1/0/2
[FW-zone-dmz] quit
[FW] firewall zone untrust
[FW-zone-untrust] add interface GigabitEthernet 1/0/1
[FW-zone-untrust] quit
```

2. Configure intrusion prevention profile **profile_ips_pc** to protect intranet users.

```
[FW] profile type ips name profile_ips_pc
[FW-profile-ips-profile_ips_pc] description profile for intranet users
[FW-profile-ips-profile_ips_pc] collect-attack-evidence enable
[FW-profile-ips-profile_ips_pc] signature-set name filter1
[FW-profile-ips-profile_ips_pc-sigset-filter1] target client
[FW-profile-ips-profile_ips_pc-sigset-filter1] severity high
[FW-profile-ips-profile_ips_pc-sigset-filter1] protocol HTTP
[FW-profile-ips-profile_ips_pc-sigset-filter1] quit
[FW-profile-ips-profile_ips_pc] quit
```

3. Create intrusion prevention profile **profile_ips_server** to protect the intranet FTP server. Configure signature 74320 as an exception signature and set the action to block.

```
[FW] profile type ips name profile_ips_server
[FW-profile-ips-profile_ips_server] description profile for intranet servers
```

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```
[FW-profile-ips-profile_ips_server] collect-attack-evidence enable [FW-profile-ips-profile_ips_server] signature-set name filter2
[FW-profile-ips-profile_ips_server-sigset-filter2] target server
[FW-profile-ips-profile_ips_server-sigset-filter2] severity high
[FW-profile-ips-profile_ips_server-sigset-filter2] protocol FTP
[FW-profile-ips-profile_ips_server-sigset-filter2] quit
[FW-profile-ips-profile_ips_server] exception ips-signature-id 74320 action block
[FW-profile-ips-profile_ips_server] quit
```

4. Commit the configuration information.

```
[FW] engine configuration commit
```

5. Configure a security policy between the Trust and Untrust zones and reference intrusion prevention profile **profile_ips_pc**.

```
[FW] security-policy
[FW-policy-security] rule name policy_sec_1
[FW-policy-security-rule-policy_sec_1] source-zone trust
[FW-policy-security-rule-policy_sec_1] destination-zone untrust
[FW-policy-security-rule-policy_sec_1] source-address 10.3.0.0 24
[FW-policy-security-rule-policy_sec_1] profile ips profile_ips_pc
[FW-policy-security-rule-policy_sec_1] action permit
[FW-policy-security-rule-policy_sec_1] quit
```

6. Configure security policies for the Trust -> DMZ and Untrust -> DMZ interzones and reference intrusion prevention profile **profile_ips_server**.

```
[FW-policy-security] rule name policy_sec_2
[FW-policy-security-rule-policy_sec_2] source-zone trust untrust
[FW-policy-security-rule-policy_sec_2] destination-zone dmz
[FW-policy-security-rule-policy_sec_2] destination-address 10.2.0.0 24
[FW-policy-security-rule-policy_sec_2] profile ips profile_ips_server
[FW-policy-security-rule-policy_sec_2] action permit
[FW-policy-security-rule-policy_sec_2] quit
[FW-policy-security] quit
```

7. Save the configuration information to upload the configuration file including the above-mentioned configurations automatically for the next startup.

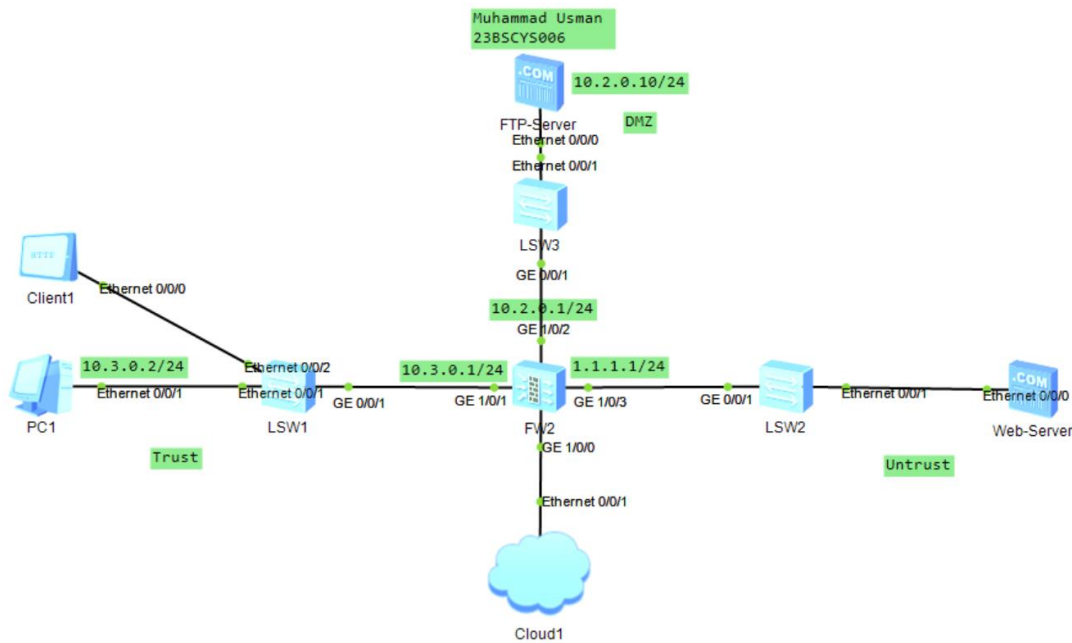
```
[FW] quit
<FW> save
```

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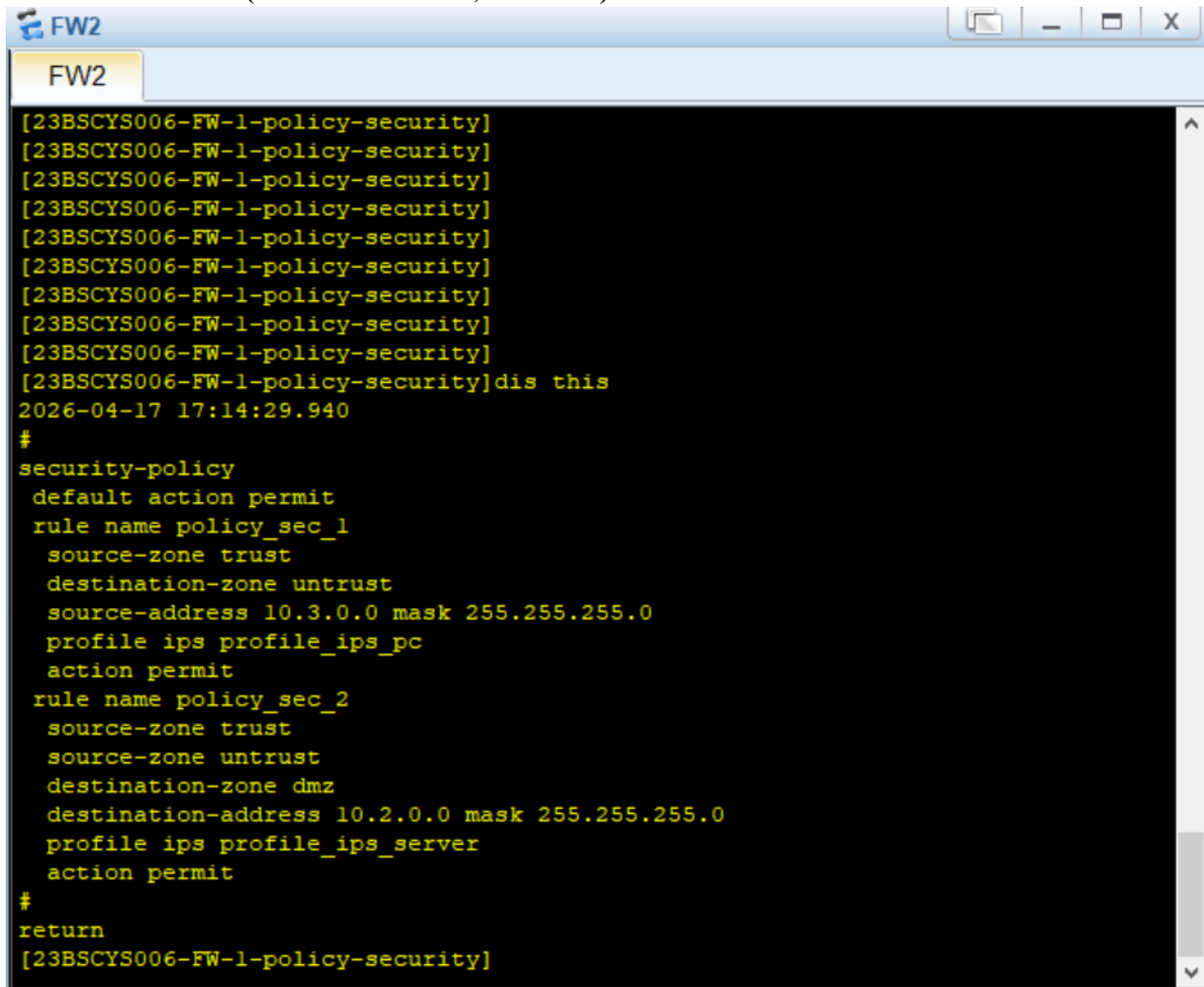
Task:

Attach Screenshots of all the topology and configuration.

Note: Attach all the screenshots of the configuration. The Firewall name should be your roll number like FW1-23BSCYS001. Upload the .topo and .cfg files on MS Teams Assignment.



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```
[23BSCYS006-FW-1-policy-security]
[23BSCYS006-FW-1-policy-security]
[23BSCYS006-FW-1-policy-security]
[23BSCYS006-FW-1-policy-security]
[23BSCYS006-FW-1-policy-security]
[23BSCYS006-FW-1-policy-security]
[23BSCYS006-FW-1-policy-security]
[23BSCYS006-FW-1-policy-security]
[23BSCYS006-FW-1-policy-security]dis this
2026-04-17 17:14:29.940
#
security-policy
  default action permit
  rule name policy_sec_1
    source-zone trust
    destination-zone untrust
    source-address 10.3.0.0 mask 255.255.255.0
    profile ips profile_ips_pc
    action permit
  rule name policy_sec_2
    source-zone trust
    source-zone untrust
    destination-zone dmz
    destination-address 10.2.0.0 mask 255.255.255.0
    profile ips profile_ips_server
    action permit
#
return
[23BSCYS006-FW-1-policy-security]
```


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```
FW2
FW2
[23BSCYS006-FW-1]dis profile ?
  type  Indicate a profile type
  <cr>

[23BSCYS006-FW-1]dis profile
2026-04-17 17:19:44.280
Profile Configurations:
Total Profiles: 18
-----
ProfileName                ProfileType  Referenced  State
-----
profile_ips_server         ips          1           committed
profile_ips_pc             ips          1           not committed
video_surveillance         ips          0           committed
strict                     ips          0           committed
web_server                 ips          0           committed
file_server                ips          0           committed
dns_server                 ips          0           committed
mail_server                ips          0           committed
inside_firewall            ips          0           committed
dmz                         ips          0           committed
outside_firewall           ips          0           committed
ids                        ips          0           committed
default                    ips          0           committed
default                    av           0           committed
default                    url-filter   0           committed
default                    aapt        0           committed
default                    casa        0           committed
default                    dns-filter   0           committed
[23BSCYS006-FW-1]
```

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FW2

POP3	0	0
NFS	0	0
SMB	0	0
IMAP	0	0
OTHER	0	0

Traffic statistic:

Total Traffic	:	638995
---------------	---	--------

Application	Traffic
HTTP	920
FTP	638075
SMTP	0
POP3	0
NFS	0
SMB	0
IMAP	0
OTHER	0

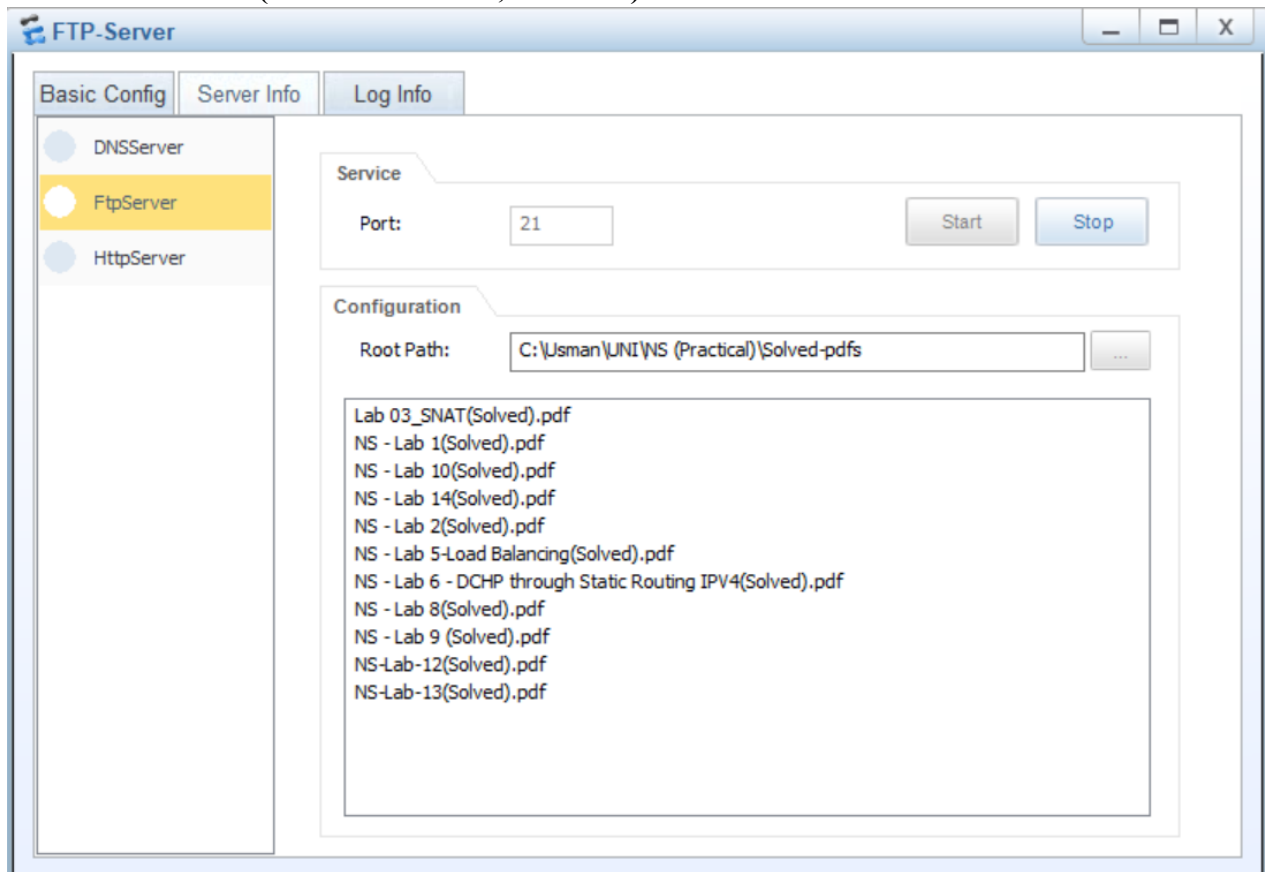
File statistic:

Total File	:	637262
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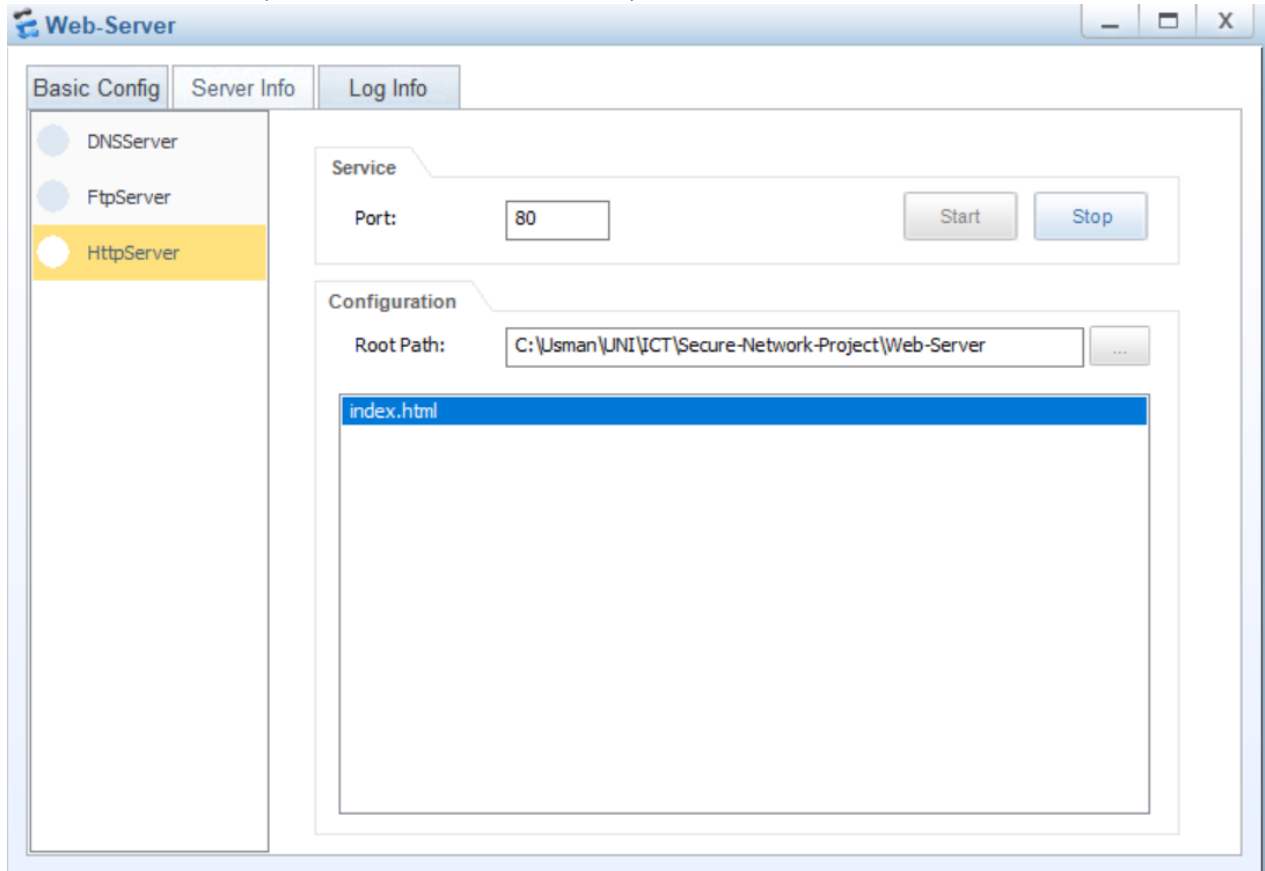
Application	File
HTTP	332
FTP	636930
SMTP	0
POP3	0
NFS	0
SMB	0
IMAP	0
OTHER	0

[23BSCYS006-FW-1]

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Client1

Basic Config | Client Info | Log Info

☒ FtpClient
☐ HttpClient

Remote Address: 10 . 2 . 0 . 10 UserName: 1
Port: 21 Password: 1

Trans Mode: ☒ PASV ☐ PORT Type: ☒ Binary ☐ ASCII ☐ Auto check Login Logout

Local Files: C:\Usman\UNI\

File Name	Size(B)
AI(practical)	
AOA	
Artifact Project	
Cyber Law and Cybe...	
Cyber Security	
DFIR	

Remote Files:

File Name	Size(B)
Lab 03_SNAT(Solved...	901460
NS - Lab 1(Solved).pdf	636930
NS - Lab 10(Solved)....	731042
NS - Lab 14(Solved)....	1046362
NS - Lab 2(Solved).pdf	599634
NS - Lab 5-Load Bala...	885067

Only display the file which is smaller than 1M

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